Case Study 5: Password Policy Validation 1. Problem statement

Develop a Python program to validate a password entered by the user. The program should check whether the password satisfies basic security requirements, such as minimum length, uppercase letters, lowercase letters, digits, and special characters. It should display whether the password is valid or invalid.

2. Objectives

- To accept a password from the user.
- To check password length and character types.
- To identify whether the password meets the required rules.
- To practice strings, loops, conditional statements, and variables in Python.

3. Assumptions

For this program, we will use the following sample rules:

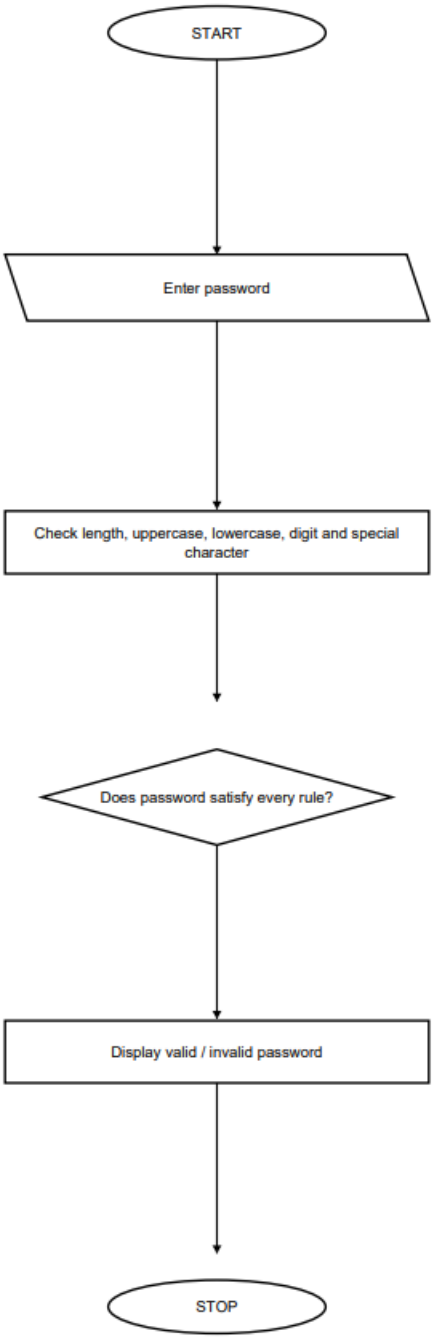
- The password must contain at least 8 characters.
- It must contain at least one uppercase letter.
- It must contain at least one lowercase letter.
- It must contain at least one digit.
- It must contain at least one special character, such as @, #, or !.
- The password is considered valid only when all five rules are satisfied.

4. Algorithm

1. Start the program.
2. Accept a password from the user.
3. Initialize variables to check uppercase letters, lowercase letters, digits, and special characters.
4. Check every character in the password.

5. Check whether the password has at least 8 characters.
6. If all requirements are satisfied, display Valid Password.
7. Otherwise, display Invalid Password.
8. Stop the program.

5. Flowchart



6. Pseudocode

START

INPUT password

SET uppercase = false SET lowercase = false SET digit = false SET special = false

FOR each character in password IF character is uppercase SET uppercase = true ELSE IF character is lowercase SET lowercase = true ELSE IF character is a digit SET digit = true ELSE SET special = true END IF END FOR

IF length of password >= 8 AND uppercase AND

lowercase AND digit AND special DISPLAY "Valid

Password" ELSE DISPLAY "Invalid Password" END IF

STOP 7. Source code

Password Policy Validation

password = input("Enter your password: ")

```
uppercase = False
lowercase = False
digit = False special =
False

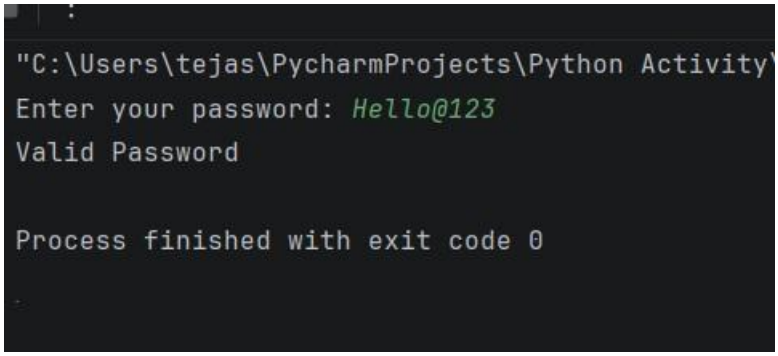
for ch in password:
if ch.isupper():
uppercase = True
elif ch.islower():
lowercase = True
elif ch.isdigit():
digit = True     else:
      special = True

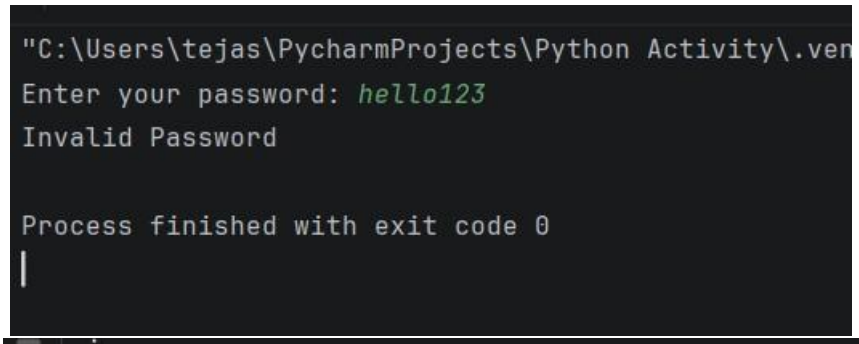
if (len(password) >= 8 and uppercase and lowercase
and digit and special):     print("Valid Password") else:
   print("Invalid Password")
```

8.Test case

Password entered	Expected output
Hello@123	Valid Password
hello123	Invalid Password
HelloWorld@	Invalid Password

9.Output







10. Conclusion

The Password Policy Validation program checks whether a password satisfies basic security requirements. It uses strings, loops, conditional statements, and built-in character-checking methods to validate the password entered by the user.